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(9A) Abstract

This transcendental phenomenological study aimed to unveil discourses on science communication in a glocal society in a Philippine graduate school classroom. Interviews were done both in virtual and limited face-to-face modes. In vivo coding technique and narrative discourse analysis were used for data analysis. Results revealed seven emerging discourses on science communication in a graduate school classroom. These are: 1) popularizing research- and evidenced-based knowledge, 2) focusing on truth and authenticity, 3) synergy of thinking skills, 4) social media engagement, 5) science communication in policymaking, 6) emancipation and empowerment through education, and peace and global understanding. The emerging discourses of Filipino graduate students reflect how they perceive science communication when positioned globally and applied locally. Science communication serves as a conduit for making science relevant and impactful to all social contexts through sound and accurate knowledge sharing and engagement of scientific and technical information. Integrating it into allied disciplines to maximize its practical application may emphasize science communication in an academic setting. Future researchers may replicate this study by exploring other methodologies and collaborating with international partner educational institutions to look into intercultural and cross-cultural constructs along this research interest.

(9B) Keywords

Discourses, classroom, graduate school, glocal society, science communication

(10) Introduction

Science communication describes various practices that transmit scientific ideas, methods, knowledge, and research to non-expert audiences in an accessible, understandable, or valuable way. Science communication audiences should not require any prior interest or educational background in science, technology, engineering, or mathematics (STEM). There is no limit to how to communicate science or who to communicate science to. It is a burgeoning multidisciplinary field that may take from various communication disciplines and styles. Science communication includes the academic study of communicating science to different publics that draws on multiple disciplines in the social sciences, humanities, and sciences. This includes research, critiques, and debates on models of science communication practiced by science communicators (Newcastle University, 2019).

Science communication describes various practices that transmit scientific ideas, methods, knowledge, and research to non-expert audiences in an accessible, understandable, or helpful way (University of Newcastle, 2014). In 2005, Dr. Ma. Theresa H. Velasco, retired dean of the College of Development Communication at the University of the Philippines in Los Baños, described science communication as fostering understanding, appreciation, and application of science and the scientific process in a manner that encourages participation by various stakeholders. Cognizance of the process of science emphasizes knowing the what, the how, and the why of science and technology communication (Velasco, 2005). In addition to learning about the latest scientific findings and discoveries, the need to understand science and the scientific process is fundamental to understanding and appreciating science and technology. It is also essential to look at science communication as a process that considers participation and interaction.

Public outreach has become an issue of growing importance for science. Scientists and scientific institutions must inform the public about potentially dangerous misconceptions or counter a continuing barrage of misinformation from numerous quarters, including commercial lobbies and fundamentalists. The communication gap between scientists and the public is due to a lack of trust, peer pressure and perception, communication uncertainty, the role of journalists, and social media.

Scientists and science communicators in the Philippines perceive local science communication efforts as poorly done and face communication challenges similar to their counterparts in developing countries. Some challenges, however, are amplified within the developing context. For example, the generally low numbers and salaries of science communicators worldwide are even lower in the Philippines. Local science communication efforts are also challenged to cater to the country's unique culture, like its many languages and conflicting attitudes to science (Navarro & McKinnon, 2020).

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The contemporary communicative landscape places new and greater responsibility on researchers and their institutions, who are increasingly active in communication with the 'end-user' and are only sometimes prepared to deal with such engagement dynamics and potential risks. In general, we could see relevant challenges for science in social research and opportunities to rethink some of the key concepts in this area (Bucchi, 2019, pp. 1-7).

Navarro and McKinnon (2020) examined the science communication challenges encountered by scientists and science communicators from Manila, Philippines. Revealed issues have been echoed in other international studies. However, challenges of accessibility and local attitudes to science were magnified within the Philippine context. These results indicate the ubiquity of specific challenges in science communication and the need for country-specific science communication frameworks. Further research on the identified challenges is needed locally and globally.

According to Bugnosen (2022), the article of Navarro and McKinnon (2020) listed several challenges faced by Filipino scientists when disseminating information to the public:

1. Mass media rarely cover science news. Science news is often shadowed by political, entertainment, or sports news and only makes headlines after events like natural disasters. Media reporting of science was found to be inadequate.

2. *Scientists need more time to communicate science.* They are more preoccupied with their research, teaching, or administrative duties and feel they should leave translating their findings to someone else.

3. *There needs to be more training in science communication.* As mentioned above, scientists have few opportunities to train as communicators, often through short workshops or fellowships.

4. *Language can be a barrier to communication activities.* The diversity of dialects present in the Philippines has implications for science communication. It is essential to speak to the stakeholders in their language to ensure they understand the message correctly or risk excluding a significant amount of the Philippine population.

5. *Science is often perceived as boring or not impacting everyday life.* This results in disinterested audiences asking how the scientific facts they are presented with will help them earn more money or feed their families. In addition, it was found that science is often communicated to those already interested in science.

Scientists need to be invited more to speak to the public. They are willing to talk about the results of their studies, but they also need someone to organize public briefings and consultations. Scientists were found to be more responsive towards outreach activities if these are clearly defined and well-coordinated by others. *There needs to be more workforce in science communication and science journalism.* The scarcity of science communicators is evident, as the same group meets and works to practice and promote science communication for years or even decades. This suggests the slow turnover of knowledge among younger communicators.

In a graduate school classroom, students have attempted to exhaust discourses on science communication which becomes underrated yet relevant in a glocal society in the age of overwhelming technological needs and advancements.

This study aimed to unveil discourses on science communication in a glocal society in a graduate school classroom.

(11) Methodology

Research Design

The study used transcendental phenomenology to examine the discourse on science communication in a graduate school classroom.

Duration and Location of the Study

This study was conducted from August to October 2022 at Capiz State University Pontevedra Campus.

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Participants of the Study

The study's participants were purposively selected ten (10) Master of Arts in Education major in English students of CAPSU Pontevedra duly enrolled in the first semester of the school year 2022-2023. They were taken in a complete enumeration.

Data Gathering Procedure

The researcher sought permission from the office of the dean of the College of Education, Arts and Science before the data gathering. The participants signed an informed consent form to ensure that ethical standards were followed and that the data generated would be used for academic exercise. The interviews were done both in a virtual and limited face-to-face manner in adherence to the learning continuity guidelines implemented by the university.

Data Analysis

Gathered data were analyzed using in vivo coding technique and narrative discourse analysis.

(12) Results and Discussions

Science Communication Discourses in a Graduate School Classroom

Figure 1 shows the discourses on science communication in a graduate school classroom. These are: 1) popularizing research- and evidenced-based knowledge; 2) focusing on truth and authenticity; 3) synergy of thinking skills; 4) social media engagement; 5) science communication in policymaking; 6) emancipation and empowerment through education; and peace and global understanding.

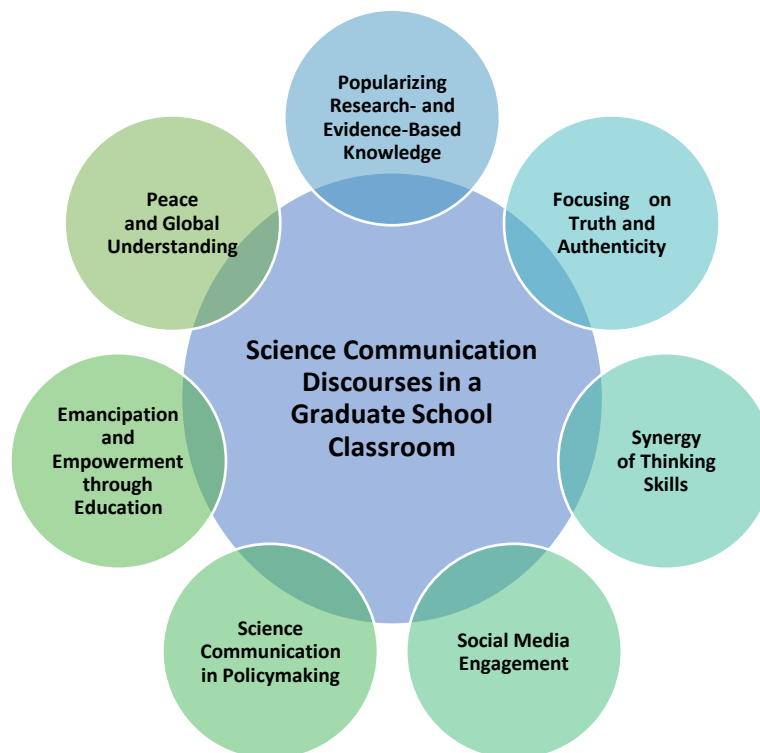


Figure 1. Science Communication Discourses in a Graduate School Classroom

Popularizing research- and evidence-based knowledge

Table 1 shows the discourse on popularizing research- and evidence-based knowledge. Scientific and technical information should be more understandable to people in a way that it will be appreciated and

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understood. Findings of Menezes et al. (2022) underscore the complexity of systemic barriers to inclusive science communication the critical need for resource sharing and network building, and the importance of evaluation frameworks. We are well on getting science communication caught up with the dramatic advances in general communication from past decades. If we all join in, the next decade will bring science communication to a whole new level, ready for many generations to come – for future leaders in science (Jansze, 2016).

Table 1. Discourse 1 – Popularizing research- and evidence-based knowledge.

Discourse	Sub-Theme	Transcript
Popularizing Research- and evidence-based knowledge	<i>Bringing science closer to people</i>	Science communication brings science closer to people. I remembered that the Department of Science and Technology had been steadfastly promoting how people would be able to appreciate the work of science.-GS 8
	<i>Promotion of evidence-based information on all platforms</i>	Communicating scientific and technical information is crucial in conscientiously positioning society's understanding of information with utmost accuracy. We need to promote evidence-based knowledge.-GS 2

Science will only be practical if communicated. It is essential to underscore how science could be promoted by all means and be far-reaching to diverse audiences. Effective science communication can empower research and innovation systems to address global challenges and put public interests at the heart of how knowledge is produced, shared, and applied. For science communication to play this mediating role effectively, we propose a more integrated and “evidence-based” approach (Jensen & Gerber, 2020).

Science communication is at a pivotal stage in its development due to the emergence of digital communication platforms that not only present new opportunities but also lead to new challenges. In this context, science communicators, including scientists, researchers, curators, journalists, and other content producers, may require new types of preparation and support to engage with multiple audiences across multiple channels (Fährnrich et al., 2021).

Focusing on truth and authenticity

Discourse on focusing on truth and authenticity is revealed in Table 2. In a generation where information is readily available, problems with discerning authenticity occur. People should not fall victim to misinformation, disinformation, and malinformation. The internet makes it easy for billions of people to access information with simple keystrokes. However, it also makes it easy to spread false information, which can have disastrous effects on both individuals and society. For this reason, it is vital to fact-check sources of information. Fact-checking is essential because misinformation can sway opinion. In turn, opinion can broadly inform actions. If actions are based on false information, one can easily make the wrong decisions. These decisions can lead to unintended consequences. For example, if you share fake news on a social media platform and people find out it's fake, it could negatively impact their opinion of your credibility. Or, the fake news you share could go viral and shape the outcome of an election (Ohio University, 2019).

Graduate students have realized how crucial agenda-setting is regarding information gatekeeping. Three aspects of agenda building are particularly noteworthy when it comes to communicating science. First, news holes (i.e., the space available for content in news outlets) has been shrinking, especially for scientific issues. In addition, the fact that major news outlets use fewer and fewer full-time science journalists (Scheufele, 2013; Dudo, Dunwoody, & Scheufele, 2011) further limits the likelihood of scientific issues making it onto the news agenda. Second, research shows that science seldom gets

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covered for the sake of science alone (Scheufele, 2014). The third aspect of agenda building that is particularly relevant to science is that the lay publics who seek information about scientific topics increasingly turn to online sources (Brossard, 2013; Brossard & Scheufele, 2013).

Table 2. Discourse 2 – Focusing on truth and authenticity.

Discourse	Sub-Theme	Transcript
Focusing on Truth and Authenticity	<i>Fact-checking</i>	Science communication has taught us to take fact-checking very seriously. In a world where information overload is just fingertips away, people could become overwhelmed in discerning the truth. We should not be victims of misinformation, disinformation, and malinformation.-GS 5
	<i>Agenda setting</i>	In Development Journalism class, we have learned that mass media can influence the importance given to various topics to what the public thinks. How we gatekeep, information may affect the decision of the people. We need to learn how issues are framed and how it is being shaped and reshaped. Right now, there is the emergence of citizen journalism. Additionally, vloggers with no formal training in journalism tend to contest the credibility of news and information from legitimate journalists.-GS 1

Synergy of thinking skills

Table 3 presents the discourse on the synergy of thinking skills. Critical thinking is needed in making judgments, especially when identifying problems.

Table 3. Discourse 3 – Synergy of thinking skills.

Discourse	Sub-Theme	Transcript
Synergy of thinking skills	<i>Critical thinking</i>	We all live in a global village anchored on a specific local setting. As world citizens, critical thinking guides us to think of problems. It is in identifying fundamental issues that we can provide real solutions. Science communication will allow us to reflect. It will enable us to think.-GS 9
	<i>Creative thinking</i>	Innovative problem-solving will foster the creation of feasible and plausible solutions to problems. It will help bridge the misunderstanding gap between nations or even in any social strata of society.-GS 3

According to Professor of Science Dr. Norman Herr, critical thinking skills can be boiled down to the following vital sequential elements: 1) Identification of premises and conclusions — Break arguments down into logical statements; 2) Clarification of arguments — Identify ambiguity in these stated assertions; 3) Establishment of facts — Search for contradictions to determine if an argument or theory

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is complete and reasonable; 4) Evaluation of logic — Use inductive or deductive reasoning to decide if conclusions drawn are adequately supported, and 5) Final evaluation — Weigh the arguments against the evidence presented. As educators, we must teach our students those critical thinking skills and practice them ourselves to objectively analyze an onslaught of information. Ideas, especially plausible-sounding philosophies, should be challenged and pass the credibility litmus test (Dominello, 2021).

Creative thinking delves into solving problems, in offering solutions. While critical thinking analyzes information and roots out the true nature and facets of problems, *creative* thinking drives progress forward when it comes to solving these problems. Exceptional creative thinkers invent new solutions to existing problems that do not rely on past or current solutions (Baker, 2014).

Social Media Engagement

The discourse on social media engagement is disclosed in Table 4. Social media has a crucial role in a global society. The extent of awareness of global issues has expanded. This prompted it to become locally relevant as it may have some effect on the ground level. The distinctive feature of global problems is the lack of a central authority for coordination and enforcement (The World Bank, 2003).

Social media platforms can be a good and bad place for debates about global challenges. The inclusiveness and equality are great, and this is unparalleled compared to offline arenas that are limited by time and space, thus highlighting the potential for social media to empower citizens, both in their role as ordinary citizens and as consumers or activists that challenge corporate behavior. On the other hand, the distinct lack of ongoing, reciprocal exchange of information or dialogue is a critical challenge, highlighting issues with using social media to debate global challenges. The lack of dialogue suggests that we must be careful about using social media platforms to discuss global challenges. Still, the use of social media to discuss global challenges is rapidly growing. Hence, we cannot disregard the importance of social media, but perhaps we can re-think their role in global discussions (The Business of Society, 2021).

Table 4. Discourse 4 – Social media engagement.

Discourse	Sub-Theme	Transcript
Social Media Engagement	<i>Taking global issues to become locally relevant</i>	Social media has a crucial role in a global society. The extent of awareness of global issues has expanded. This prompted it to become locally relevant as it may affect the ground level.- GS 10
	<i>Effective communication of science over social media</i>	Scientific knowledge or results are communicated effectively if social media is used well. If social media is used correctly, conflicts over beliefs and values will be minimized, if not entirely avoided. It is still a long way, but it will work.-GS 6
	<i>Understanding competing messages and sources of information</i>	It is not an issue of which social media platform is better for understanding messages or information. It is a matter of addressing the problem of making competing messages and sources of information better understood and clearly understood.-GS 4

The Internet and social media tools have created new opportunities for open science, including communicating more effectively and sharing research data. Drawing on evidence from interviews and a survey of academics in the United Kingdom, our research suggests that most scholars recognized the value and importance of more open science communication and data sharing, but many had concerns about the potential risks. A small group termed super users frequently communicated updates of their

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ongoing research. It is clear that there are increasing opportunities for more open science and public engagement, but challenges remain (Zhu & Purdam, 2017).

Science communication in policymaking

Table 5 shows the discourse on science communication in policymaking. Science communication allows policymakers to be guided when making decisions.

Table 5. Discourse 5 – Science communication in policymaking.

Discourse	Sub-Theme	Transcript
<i>Science Communication in Policymaking</i>	<i>The scientific and technical information in formal policy processes</i>	Science communication gives policymakers adequate and necessary scientific and technical information to substantiate formal policy processes. This is vital in making the policies proactive to serve their very purpose.-GS 7

Scientific and technical information is increasingly used in National Research Council reports, and its use is associated with a variety of report characteristics, aspects of the public policy environment, committee characteristics, and the origin of the request. On the other hand, scientific and technical information does not make National Research Council reports more likely to be used by Congress (Youtie et al., 2017, p. 108).

Emancipation and empowerment through education

The discourse on emancipation and empowerment through education is reflected in Table 6. Sound decisions are based on science-based evidence. This evidence will allow people to have options or alternatives. Digital divides, in particular, are a significant structural barrier, and community literacy and building science literacy in media and science communication are promising opportunities (Howell & Brossard, 2021).

Table 6. Discourse 6 – Emancipation and empowerment through education.

Discourse	Sub-Theme	Transcript
<i>Emancipation and Empowerment through Education</i>	<i>Science communication in making informed choices and decisions</i>	Science-based information must provide informed choices and decisions to people. This is the core of science communication, allowing people to figure it out themselves. We want people to be empowered. They will be emancipated.-GS 2
	<i>Role of academic institutions in propelling science communication</i>	Educational institutions, including ours, has a role in making science communication felt. The academe can propel science communication as a catalyst for positive change.-GS 7

The ecosystem of science communication is diversifying. Science communication is becoming professionalized. Science communication is no longer the embattled preserve of a few 'visible scientists'. It is a domain populated by a wide range of full-time communicators (alongside many enthusiastic volunteers), whether those are scientists who have turned to communication full-time, communication professionals from other domains (such as PR), or a growing cadre of individuals specifically (Davies & Horst, 2016).

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While many palaeoscientists did express firm commitments to science communication, they had previous experience with time constraints and conflicts with other academic responsibilities. Therefore, palaeoscientists and their institutions would benefit from viewing science communication as a valuable and formally rewardable activity within the scholarship of sharing knowledge, which also contributes to other aspects of a successful academic career (Barbolini, 2022).

Peace and global understanding

Table 7 presents the discourse on peace and global understanding. Being good communicators, in general, will help scientists to be better science communicators to 1) help the public understand science as part of their natural lives, 2) see not only the importance of science and its source of pleasure and wonder but also to be able to make decisions about it as citizens, policymakers, funders, etc., 3) help in educating citizenry, concerned about the threats facing our planet to better shape the direction of political and policy decisions, 4) because how scientists communicate this information may have measurable conservation impacts on the future of our planet (Jucan & Jucan, 2014).

Table 7. Discourse 7 – Peace and global understanding.

Discourse	Sub-Theme	Transcript
<i>Peace and global understanding</i>	<i>Impact for a better world</i>	For me, science communication changes our knowledge, attitude, and practice in any aspect of society – political, economic, and all others that directly impact our lives. We need it for a better world. World peace is said to be impossibly attained. However, when we try to embrace cultural diversity and understanding, we develop a sense of sensitivity toward others. Peace will prevail.-GS 5
	<i>Preparing learners for future-proofing</i>	Learners must be prepared in these times of uncertainty. Science communication will help us realize the need for future-proofing. We have to anticipate the future. We plan and re-plan to be successful in the future, even if the situation changes.-GS 8

The current multiple revolutions in science and technology have an immense impact on society today and affect the future of humanity and the Earth. Science and society look for and need each other. Scientists have an ethical obligation to the public to produce factual, intelligible, timely information and to account for their stewardship of the public funds used to support their work. Communications are adequate if they reach people with the information they need in a form they can use (Jucan & Jucan, 2014).

The world is changing, and so are our instructional practices. Our moral obligation is to equip this generation with the content and abilities to handle an unpredictable future and thrive. Doing so may be our most important educational task. We should focus on providing education that recognizes the relevance and understanding of global issues in their social, political, cultural, economic, and environmental dimensions. Therefore, a learner's knowledge and skills should be developed to secure a more just, peaceful, tolerant, inclusive, secure, and sustainable world.

Being future-proofed means standing the test of time in the face of changing contexts and challenges. University policies are most effective when they are designed to withstand and accommodate future changes in the educational landscape. These policies reflect agreed-upon principles that will withstand social and political changes (Naidu & Roberts, 2018)

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(13) Conclusions and Recommendations

Conclusions

The emerging discourses of Filipino graduate students reflect how they perceive science communication when positioned globally and applied locally.

Science communication serves as a conduit for making science relevant and impactful to all social contexts through sound and accurate knowledge sharing and engagement of scientific and technical information.

Recommendations

Science communication may be emphasized in an academic setting by integrating it into allied disciplines to maximize its practical application.

This study may be replicated by exploring other methodologies and collaborating with international partner educational institutions to look into intercultural and cross-cultural constructs along this research interest.

(14) Output

(15) Technology Transfer Activity Conducted

(17) Acknowledgments

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